Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 11: Node Voltage Analysis

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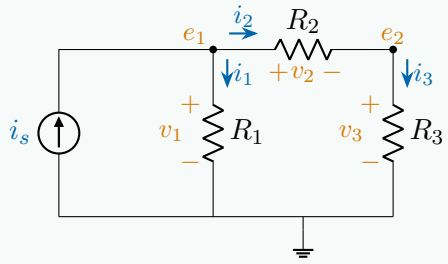
Colour code used in these notes and in the slides		
current i	voltage v, node voltage e	supernode, nodes Ⓢ

Motivation. An LTI resistive circuit can be solved in two ways: by transforming it into a simpler circuit, or by writing its equations as it is. Transforming works for many circuits, but not for all. This lecture gives a systematic way to write the equations of *any* LTI resistive circuit: **node voltage analysis**. Its unknowns are only the $n-1$ node voltages, instead of the $2b$ branch voltages and currents. Mesh current analysis, the same idea with KCL and KVL swapped, comes later.

1. The idea: node voltages as the unknowns

The circuit below has 4 branches, so 4 branch voltages and 4 branch currents: 8 unknowns. With node voltages, only 2 unknowns are left. Three steps show how.

A small example



Step 1: branch voltages. With the bottom node as the ground, $v_1 = e_1$, $v_2 = e_1 - e_2$, $v_3 = e_2$. KVL around the loop, $v_1 = v_2 + v_3$, holds automatically: $e_1 = (e_1 - e_2) + e_2$.
Step 2: branch currents. Ohm's law gives $i_1 = e_1/R_1$, $i_2 = (e_1 - e_2)/R_2$, $i_3 = e_2/R_3$.
Step 3: KCL. At nodes 1 and 2:

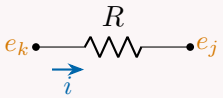
$$-i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0, \qquad -\frac{e_1 - e_2}{R_2} + \frac{e_2}{R_3} = 0.$$

Two equations in e_1, e_2; then Steps 1 and 2 give every branch voltage and current.

In general, $\mathbf{v} = A^T\mathbf{e}$ (KVL holds automatically), the element laws give $\mathbf{i}$, and KCL $A\mathbf{i} = \mathbf{0}$ gives $n-1$ equations in the $n-1$ node voltages.

Node voltage analysis: the procedure

1. Choose a **reference node** (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.
3. Write **KCL** at every node except the ground, with each current written in terms of node voltages.



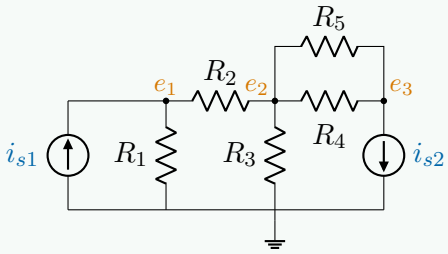
$$i = \frac{e_k - e_j}{R} = G\,(e_k - e_j)$$

The current *leaving* node k through R; with the other end at the ground, it is e_k/R.

2. Resistors and current sources

Start with circuits whose elements are only resistors and current sources. Every branch current is then either known (a source) or a conductance times a difference of node voltages. The circuit of Example 11.1 has $b = 7$ branches, so $2b = 14$ branch voltages and currents. With the bottom node as the ground, only $n - 1 = 3$ node voltages are unknown.

Example 11.1 — only current sources: KCL at nodes 1, 2 and 3



node 1: $$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

node 2: $$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} + \frac{e_2 - e_3}{R_5} = 0$$

node 3: $$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_2}{R_5} + i_{s2} = 0$$

Every term is a current leaving the node; the source i_{s1} points into node 1, so it enters with a minus sign. With $G_k = 1/R_k$ and the terms collected, the three equations of Example 11.1 become

$$\underbrace{\begin{bmatrix} G_1 + G_2 & -G_2 & 0 \\ -G_2 & G_2 + G_3 + G_4 + G_5 & -(G_4 + G_5) \\ 0 & -(G_4 + G_5) & G_4 + G_5 \end{bmatrix}}_{Y_n} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_s}.$$

Y_n is the **node admittance matrix**, and $Y_n\mathbf{e} = \mathbf{i}_s$ are the **node equations**. For a circuit of resistors and current sources, they can be written down directly.

Writing Y_n by inspection

- **Diagonal** entry (k, k): the sum of the conductances connected to node k.
- **Off-diagonal** entry (k, j): minus the sum of the conductances between nodes k and j.
- **Right side**, row k: the sum of the source currents entering node k.

So Y_n is **symmetric**: the entry (k, j) equals the entry (j, k).

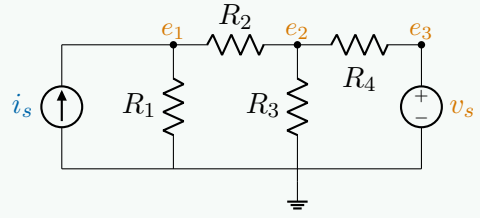
Why is Y_n symmetric? Split the columns of A into the resistor branches and the current source branches, $A = [\,A_R \;\; A_S\,]$. KCL reads $A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$. The resistors obey $\mathbf{i}_R = G\,\mathbf{v}_R$ with $G = \mathrm{diag}(G_1, \ldots, G_5)$, and KVL gives $\mathbf{v}_R = A_R^T\mathbf{e}$. Together,

$$\underbrace{A_R\,G\,A_R^T}_{Y_n}\,\mathbf{e} = \underbrace{-A_S\mathbf{i}_S}_{\mathbf{i}_s}, \qquad Y_n^T = A_R\,G^T\,A_R^T = A_R\,G\,A_R^T = Y_n,$$

since G is diagonal. $A_S\mathbf{i}_S$ adds up the source currents leaving each node, so $-A_S\mathbf{i}_S$ adds up those entering it.

3. A voltage source to the ground

Example 11.2 — a voltage source between node 3 and the ground



The unknowns are e_1, e_2, e_3, so three equations are needed. KCL at nodes 1 and 2:

node 1: $$-i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

node 2: $$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Question. What does KCL at node 3 give?

Answer. The current through v_s is unknown, and it cannot be written with node voltages. So KCL at node 3 gives no useful equation in e_1, e_2, e_3. But the source gives $e_3 = v_s$: this is the third equation.

Remark: a voltage source to the ground

If an independent voltage source v_s is connected between the ground and node k, then $e_k = v_s$ is known. So e_k is not an unknown, and KCL is not written at node k.

Example 11.2, revisited

The unknowns are only e_1 and e_2. Put $e_3 = v_s$ into the two KCL equations:

$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}.$$

The known v_s moves to the right side: v_s/R_4 acts like a current into node 2.

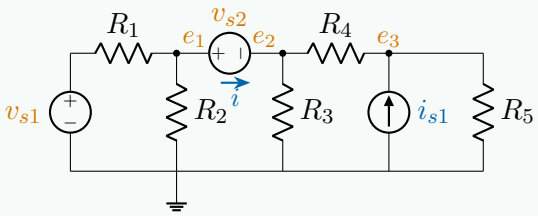
4. A voltage source between two nodes: modified node analysis

If a voltage source (independent or dependent) lies between two nodes and neither is the ground, KCL there needs its current. Its law fixes its voltage, not its current, so this current cannot be written with node voltages. **Modified node analysis** offers two ways out.

Method I

Let the unknown current i through the voltage source be one more unknown. Write KCL at every node except the ground, and add the law of the source as a **constraint equation**.

Example 11.3 — Method I



The top of v_{s1} is at the known voltage v_{s1}; v_{s2} is between nodes 1 and 2. Unknowns: e_1, e_2, e_3 and i.

node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

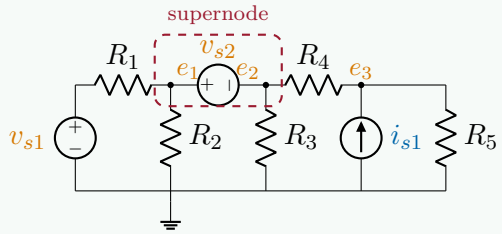
constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Method II: the supernode

Write KCL as if the two nodes of the voltage source were a single node, a **supernode**. Add the law of the source as a constraint equation.

Example 11.3, revisited — Method II, three unknowns



supernode: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

constraint: $e_1 - e_2 = v_{s2}$

Why does it work? Add the KCL equations of nodes 1 and 2 in Method I: i leaves node 1 and enters node 2, so it cancels. The sum is KCL around the supernode. The current i is no longer needed. In matrix form, with three unknowns and three equations:

$$\begin{bmatrix} G_1+G_2 & G_3+G_4 & -G_4 \\ 0 & -G_4 & G_4+G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}.$$

The matrix is no longer symmetric.

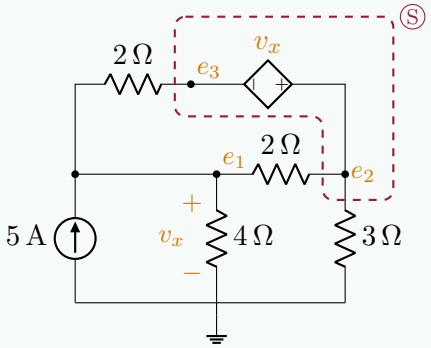
	Method I	**Method II**
extra unknown	the current i of the source	none
KCL written	at every non-ground node	at the supernode and the other nodes
extra equation	the constraint (law of the source)	the same constraint
size in Example 11.3	4×4	3×3
the current i	part of the solution	from a KCL afterwards

Both methods give the same node voltages.

5. Dependent sources

A dependent source is treated like an independent source of the same kind. Its control variable is written with node voltages; this gives one more constraint equation.

Example 11.4 — a dependent voltage source



Question. Find the power absorbed by the $3\,\Omega$ resistor.

The source v_x is between nodes 3 and 2, and neither is the ground. So use a supernode Ⓢ around nodes 2 and 3. The unknowns are e_1, e_2, e_3.

(1) node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$

(2) Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$

Constraint. The source gives $e_2 - e_3 = v_x$, and its control v_x is the voltage of the $4\,\Omega$, so $v_x = e_1$:

$$e_2 - e_3 = e_1. \qquad (3)$$

Solving Example 11.4. Put $e_1 = e_2 - e_3$ into (1) and (2):

$$\tfrac{5}{4}e_1 - \tfrac{1}{2}e_2 - \tfrac{1}{2}e_3 = 5 \;\Rightarrow\; \tfrac{3}{4}e_2 - \tfrac{7}{4}e_3 = 5, \qquad -e_1 + \tfrac{5}{6}e_2 + \tfrac{1}{2}e_3 = 0 \;\Rightarrow\; e_3 = \tfrac{1}{9}e_2.$$

Then $\frac{3}{4}e_2 - \frac{7}{36}e_2 = \frac{20}{36}e_2 = 5$, so $e_2 = 9\,\text{V}$, $e_3 = 1\,\text{V}$ and $e_1 = 8\,\text{V}$. The $3\,\Omega$ is between node 2 and the ground:

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\text{W}.$$

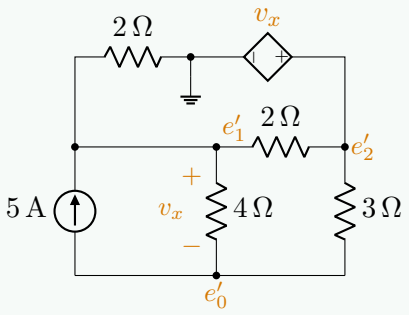
Remark: a dependent voltage source to the ground

If a dependent voltage source is connected between node k and the ground, do not write KCL at node k: its current is not known. Write the constraint that the dependent source gives instead.

6. Choosing a different ground

Any node can be the ground. Changing it shifts every node voltage by the same amount, so the branch voltages, the currents and the powers do not change.

Example 11.5 — Example 11.4 with node 3 as the ground



Question. Redo Example 11.4 with node 3 as the ground. Does the answer change?
Now v_x is between node 2 and the ground: no supernode is needed, and by the remark above, KCL is not written at node 2. The unknowns are e_1', e_2' and e_0', the voltage of the old ground.

node 1: $$-5+\frac{e_1'}{2}+\frac{e_1'-e_0'}{4}+\frac{e_1'-e_2'}{2}=0$$

node 0: $$5+\frac{e_0'-e_1'}{4}+\frac{e_0'-e_2'}{3}=0$$

constraint: $$e_2'=v_x=e_1'-e_0'$$

Solving Example 11.5. Put the constraint into the two KCL equations: $3e_1'+e_0'=20$ and $-7e_1'+11e_0'=-60$. So $e_1'=7\,\text{V}$, $e_0'=-1\,\text{V}$, $e_2'=8\,\text{V}$, and

$$P_{3\Omega}=\frac{(e_2'-e_0')^2}{3}=\frac{9^2}{3}=27\,\text{W},\quad \text{the same.}$$

Every node voltage dropped by $e_3=1\,\text{V}$: $8\to7$, $9\to8$, $0\to-1$.

Tip. Put the ground at one end of a voltage source. Then its other node voltage is known, or a supernode is avoided.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e}=\mathbf{i}_s$, written by inspection; $Y_n=A_RGA_R^T$ is symmetric
a voltage source from the ground to node k	$e_k=v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint; Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint equation
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages are found: $\mathbf{v}=A^T\mathbf{e}$, then the branch currents from the element laws.